

Milestone 1

Problem Definition & Literature Review [October 3]

1. Define the objectives of the project:

Objective: To build a Clothing Assistant.

A chatbot designed to answer questions from store policy, previous purchases of customers, and suggest outfits based on query given and customer preferences.

Part A: Store policy + Customer help desk

The chatbot will answer questions from store policies like what is the return policy, upcoming discount info, and so on. Additionally the customer can select a previously bought item and ask something like whether I can return or replace the product. Now the agent will check the return or replacement policy of that specific product and reply accordingly.

Part B: AI stylist

- The agent will be fed context about the customer preferences (liked outfits, size, etc). Without any query given the agent should concentrate on these preferences and suggest clothes. An additional advanced option could be suggesting clothes based on the skin tone of the customer if shared.
- Initially to get customer preferences and serve well a quiz will be designed.
- Given a query the agent should consider both customer preferences and query given and suggest clothes. (An example query would be 'suggest me an outfit for a graduation party')
- To prevent hallucinations the agent will only suggest products from the company catalogue. Also the agent should stick to the job of answering from company policy or suggesting outfits.
- The agent should have knowledge about fashion like which color shirt matches with which color pants, on which occasions what outfit is appropriate, etc,...

2. Review of existing solutions, baselines, and benchmarks:

a. Existing solutions -

- **H&M “Virtual Stylist”**, experimental assistants / chatbots for style recommendations, virtual try-on etc.
- **Klarna’s “Shop-with-me / AI Style advisor”**, Style recommendations, curated product browsing, conversational interface in app / web.
- **Yesplz AI**. Quiz based model where a set of questions is framed and products are shown based on the answers. A similar kind of approach can be used for filtering the suggestions.
- **PairFashionExplanation** — Explains how prompt engineering can be used to get matching pairs of clothes. The dataset used is fashion VC which has image + description.

b. Current baselines and benchmarks -

- **Fashion-IQ (conversational / relative feedback image retrieval)** — a public dataset created to study dialog-style fashion image retrieval. **Recall** ($R@10/R@50$ based metrics) ranging from ~39% to ~66%.

We could use **Avg Recall@10 / Avg Recall@50** as the baseline metric for retrieval tasks.

[The **Recall@10** is the number of relevant items in the top 10 results divided by the total number of relevant items in the entire dataset for a given query.]

3. Data Sourcing, Licensing, and Embeddings:

For the **catalog**, we are considering a dataset containing all product details along with images, so that when a customer requests a particular item, the AI can search the dataset and display the corresponding product image. The initial thought process is we will embed the catalog and store it in a vector database for retrieval.

For **policies**, we will refer to the guidelines of major online clothing brands like Myntra and Bewakoof as baseline references, and then customize them to generate a comprehensive internal policy document for the chatbot.

Both the FAQ's and policy documents will be converted into embeddings using models like Sentence-BERT (for text), and store in a vector database for fast and accurate retrieval during chatbot queries.

4. Gaps and Opportunities:

- Existing solutions relied on multiple external websites instead of a single catalogue, causing agents to suggest clothes outside of our catalogue. Whereas our AI will use only the company's internal product catalogue with all item details and images. Recommendations will be grounded in this dataset, ensuring items are available and from our catalogue. This prevents promoting third-party products.
- **The agents did not stick to the customer request** (when asked for a pant, the agents suggested other products too). So we will implement query-based filtering in the retrieval pipeline, using product metadata and embeddings to ensure the AI only selects items that match the customer's specific request.
- **The agents did not have much fashion sense.** Not always relevant products were shown. For example, when asked to suggest a pant for an already available red shirt, the agent did not pair a proper pant. Actually the pants suggested didn't even take into account the red shirt. So we will create sets of rules for a GPT, so that outfit suggestions are context-aware and items like pants are paired appropriately with the user's existing clothing.
- **We didn't find any agents which requested for size** and filtered based on that too. The agents suggested a clothe and after the customer selects the product it showed sizes available. We will include size information in the product embeddings and apply size-based filtering during retrieval, so the AI only recommends items available in the user's preferred size.

- **The chat history is not maintained properly.** For example, we requested an outfit for a business meeting, and the agent suggested some blazers. Next we gave a prompt 'need a blue one'. Here the bot forgot about the occasion 'business meeting' and suggested even T-shirts. We will implement a vector-based conversational memory to retain context, ensuring follow-up queries consider previous prompts like occasion or item type.

References:

1. Yesplz: <https://yesplz.ai/product-discovery/ai-stylist>
2. Fashion IQ: <https://arxiv.org/pdf/1905.12794>
3. Pair Fashion Explanation:
<https://github.com/wangyu-ustc/PairFashionExplanation/blob/main/README.md>
4. Fashion VC:
<https://drive.google.com/file/d/1d72E3p4w280-vdCKfXtXZLMULpIScRRR/view>